Hyperspectral Remote Sensing Object Detection via Cross-Domain Learning From Visible Images

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***Abstract*—Due to the high cost of data acquisition and annotation, it is challenging to obtain large-scale training data for hyperspectral remote sensing images (HRSIs), making it difficult to train object detectors directly. In contrast, visible-light remote sensing images (VLRSIs) are abundant and well-annotated. Hence, cross-domain object detection from VLRSI to HRSI (VLRSI2HRSI) provides a feasible solution for HRSI detection. However, existing cross-domain detection methods are developed for natural images. They neglect the shared spatial characteristics in the remote sensing scenario and fail to model the spectral properties of HRSIs. As a result, directly applying them to VLRSI2HRSI leads to significant performance degradation. To address this issue, we propose a visible-light-to-hyperspectral cross-domain detection network (VHCDN). We first observe that reconstructed residual representations in both VLRSI and HRSI suppress background and emphasize foreground objects. Based on this observation, we introduce a residual extracting module (REM) to extract domain-shared spatial features. Furthermore, since objects of the same semantic class may consist of different materials and exhibit spectral variations under varying illumination, we design a spectral saliency head (SSH) that models spectral uncertainty. By introducing one HRSI into the source-domain VLRSI datasets to guide the generation of spectral features, the SSH is trained with a spectral heteroscedastic loss and a spectral uncertainty loss, enabling the uncertainty learned in the labeled source domain to be stably transferred to the unlabeled target domain. Experiments on three cross-domain settings against seven representative methods demonstrate that VHCDN achieves state-of-the-art performance, while ablation studies verify the effectiveness of each proposed module.**

***Index Terms*—Cross domain, hyperspectral image, object detection.**

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I. Introduction

HYPERSPECTRAL remote sensing images (HRSIs) not only capture spatial information in two dimensions but also record the spectral signature of each pixel along a third dimension, thereby providing richer and more discriminative information than visible-light remote sensing images (VLRSIs). Due to this advantage, HRSIs have attracted increasing attention from the remote sensing community [1], [2]. Among various downstream tasks [3], object detection [4], [5], [6], as a fundamental and representative task [7], [8], has been extensively applied in numerous scenarios including national defense and security [9], resource exploration [10], and environmental monitoring [11], [12]. Due to the high cost of collecting and annotating HRSIs, it is challenging to acquire large-scale datasets for training, which limits the generalization capability of HRSI-based object detection. In contrast, the cost of acquiring VLRSIs is relatively low, and a large amount of labeled data is already available. Therefore, cross-domain object detection from VLRSIs to HRSIs becomes a promising solution for achieving effective object detection in the hyperspectral domain, which is the main focus of this article.

Existing cross-domain object detection methods are primarily designed for common visible light images and can be broadly categorized into two types: adversarial learning-based methods and self-training-based methods. Adversarial learning-based approaches aim to learn domain-invariant representations by reducing the feature discrepancy between the source and target domains. In contrast, self-training-based approaches first train a detector on the source domain and then generate reliable pseudo-labels for the target domain to facilitate adaptation through target-domain training. However, these methods fail to account for the domain-shared spatial characteristics in the remote sensing field and neglect the modeling of spectral features. Consequently, directly applying them to the VLRSI–HRSI cross-domain object detection task leads to significant performance degradation. Therefore, it is necessary to develop a novel cross-domain object detection framework that jointly considers the domain-shared spatial representations and the spectral feature modeling between VLRSIs and HRSIs.

To this end, we design a visible-light-to-hyperspectral cross-domain object detection network. The motivation behind visible-light-to-hyperspectral cross-domain detection network (VHCDN) stems from two key observations. First, we observe that in remote sensing images, whether in HRSI or VLRSI datasets, the proportion of objects is much smaller than that

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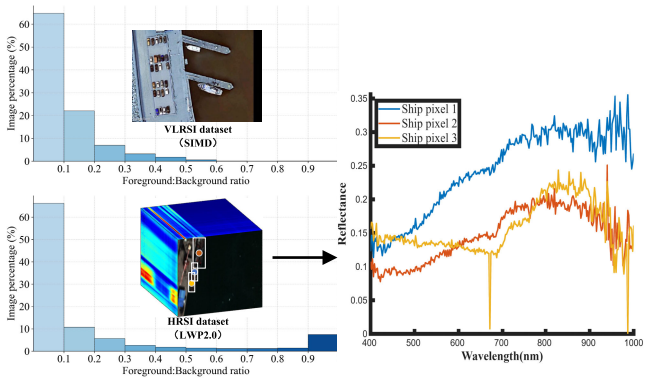


Fig. 1. Observation 1: as the object ratio in most remote sensing datasets is smaller than the background ratio, the reconstruction process focuses on the larger background area. Therefore, the residual features of the VLRSI domain and the HRSI domain have common spatial features to highlight the object of interest. Therefore, aligning the residual features helps to unify the spatial features between domains. Observation 2: spectral features of the same semantic category have spectral uncertainty due to different composition materials or environmental changes. Therefore, this uncertainty should be considered in cross-domain modeling.

of the background. During the reconstruction process, objects tend to be less accurately reconstructed, making them more prominent in the residual features. Therefore, the residual features can be regarded as domain-shared spatial representations between HRSIs and VLRSIs, as illustrated in the left of Fig. 1. This phenomenon is statistically supported by existing hyperspectral remote sensing benchmarks. For example, in the LWP2.0 dataset, object instances are sparsely distributed across 224×224 scenes, and the effective object-pixel ratio is significantly lower than that of background pixels. This structural sparsity motivates the reconstruction-based modeling strategy.

Second, we notice that features belonging to the same semantic category exhibit spectral uncertainty due to differences in material composition and environmental conditions, as shown in the right of Fig. 1. Hence, it is necessary to model the spectral uncertainty. In real hyperspectral imaging systems, spectral variability arises from multiple physical factors, including material composition heterogeneity, illumination variation, atmospheric scattering, sensor noise, and subpixel mixing effects. Even within the same semantic class, coatings, aging conditions, and shadowing may cause noticeable spectral dispersion.

Based on Observation 1, we propose a residual extracting module (REM), which simultaneously extracts residuals from both the source-domain VLRSIs and the target-domain HRSIs in a self-reconstruction manner, thereby indirectly aligning the domain-shared spatial representations. Based on Observation 2, we introduce a spectral saliency head (SSH). To enable the VLRSIs domain to generate spectral representations similar to those of the target domain guided by prior spectra, we design a spectral calibration module (SCM) based on radial basis function (RBF) modeling. Furthermore, to jointly account for spectral uncertainty in both source and target domains, we impose a spectral heteroscedasticity loss and a pseudo-confidence-guided binary cross-entropy loss on the SSH, ensuring that the spectral uncertainty learned in the labeled source domain can be transferred to the unlabeled target domain.

In summary, this article presents the VHCDN for the VLRSIs to HRSIs cross-domain object detection task. To extract domain-shared spatial features between VLRSIs and HRSIs, a REM is proposed. This module reconstructs both source and target domains in a self-reconstruction manner to obtain residuals, which serve as domain-shared spatial features. To model spectral uncertainty, an SSH is designed. In the source section, VLRSIs are passed through the SCM, where, guided by the prior spectra of one precollected HRSI, the SCM generates spectral curves similar to those of the target domain based on an RBF modeling. To align spectral uncertainty across domains, the SSH is further constrained by a spectral heteroscedastic loss and a spectral uncertainty loss, enabling the spectral uncertainty learned from the labeled source domain to be effectively transferred to the unlabeled target domain.

The main contributions of this article are summarized as follows.

1) We propose a cross-domain object detection method between VLRSIs and HRSIs to address the challenge of object detection in HRSIs under limited HRSI data availability.
2) We propose a residual extracting module to address the spatial-domain alignment problem between VLRSIs and HRSIs.
3) We propose an SSH to model spectral uncertainty.
4) We conduct comparative and ablation experiments to verify the effectiveness of the proposed method and modules.

The rest of this article is organized as follows. Section II reviews the related studies, and Section III details the proposed approach, including the REM and SSH. Section IV reports the experimental results, and Section V concludes this article.

II. Related Work

It is worth noting that existing cross-domain object detection methods are not specifically designed for VLRSI–HRSI scenarios. Therefore, in this section, we review both general cross-domain object detection methods for visible light images and existing HRSI object detection approaches.

A. HRSI Object Detection Approaches

According to whether prior spectral information is available, existing HRSI object detection methods can be categorized into hyperspectral anomaly detection (HAD) methods, which do not require prior spectra, and hyperspectral target detection (HTD) methods, which rely on prior spectral signatures. The former identifies anomaly pixels by locating pixels that exhibit spectral inconsistencies with their surrounding neighborhood, whereas the latter detects objects by searching for pixels whose spectral curves are similar to the given prior spectra.

1) Hyperspectral Anomaly Detection: According to their methodological principles, HAD approaches can be broadly divided into model-driven and data-driven methods. Model-driven methods rely heavily on prior assumptions—for

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example, the classical RXD [13] detector assumes that background pixels follow a Gaussian distribution and identifies anomaly pixels as those that significantly deviate from this distribution. In contrast, data-driven methods directly reconstruct the HRSI; since anomaly pixels typically occupy a much smaller proportion than background pixels, the reconstruction residuals are utilized as anomaly maps [14] for object detection.

The autoencoder (AE) is an unsupervised deep learning technique that learns the latent representations of data by minimizing the reconstruction error between the input and its output. In HAD, the AE-based method proposed by Bati et al. [15] follows this principle, identifying an anomaly by evaluating the reconstruction error. To overcome the limitations of traditional supervised learning methods that rely on labeled samples, Siamese networks reformulate the HAD as a similarity measurement task, determining anomalies based on the similarity between pixel pairs. The Siamese-based detection framework [16] not only enhances the model's transferability but also reduces its dependence on large-scale annotated data. Subsequently, Wang et al. [4] argue that jointly fitting both background and anomalous pixels is suboptimal, and therefore developed a reconstruction network based on a dual-window mechanism. By introducing a blind zone at the center of the receptive field, the network focuses on reconstructing the central region while suppressing background interference, thereby significantly improving HAD performance.

2) Hyperspectral Target Detection: The origin of HTD can be traced back to the fundamental concept of spectral similarity. For example, the spectral angle mapper (SAM) [17] measures the angular difference between a pixel and a prior spectrum to obtain a matching score. Later, the constrained energy minimization (CEM) [18] method assumes that the target spectrum maintains a constant intensity and emphasizes target pixels indirectly by minimizing the overall energy of the HRSI, thereby achieving more robust detection performance [19].

To introduce deep learning models into the field of HTD, Du and Li [20] reformulate the task as a binary classification problem, where target pixels are treated as positive samples and background pixels as negative samples. A neural network is then trained to determine the class membership of each pixel. To further capture global spectral–spatial features, Qin et al. [21] introduce the Vision Transformer (ViT) architecture and propose a joint spectral–spatial hyperspectral matching detection method.

To fully exploit prior spectral information, several methods [22], [23] reformulate the HTD as a similarity matching problem between the spectrum of a test pixel and the known spectral prior. In these approaches, a pair of pixels belonging to the same category (either both target or both background pixels) is treated as a positive sample, while a pair of pixels from different categories (one target and one background) forms a negative sample. A neural network is then trained to determine whether the input pixel pair belongs to the same class. However, the imbalance between positive and negative samples limits the performance of such methods.

To address this issue, many researchers have attempted to generate synthetic samples based on the available data. For example, Freitas et al. [24] generate target samples using an AE and then employ a linear prediction strategy to obtain corresponding background samples. Later, Zhao et al. [25] point out that most existing HTD methods largely ignore the structural information of HRSI, which restricts detection accuracy. To overcome this limitation, they propose an HTD based on tensor-guided adaptive reconstruction combined with global–local feature fusion (TRGLF) in a cascaded framework.

Despite the remarkable progress of the aforementioned methods in hyperspectral object detection, they all perform pixel-level detection within individual images. This is primarily due to the extreme scarcity of large-scale HRSI datasets, which hinders the generalization of object-level detection across multiple scenes.

Meanwhile, object detection in optical remote sensing imagery has also attracted considerable attention in recent years. For instance, Zhang et al. [26] propose a high-resolution feature generator to enhance small-ship detection in optical remote sensing images, while Guo et al. [27] introduce a multi-instance mining framework with dynamic localization for weakly supervised object detection in remote sensing imagery. These studies provide useful insights for the development of object detection frameworks in remote sensing. In parallel with these developments, Li et al. [28] designed a feature augmentation module and an oriented feature alignment module to achieve a better balance between classification and regression tasks, significantly improving detection performance.

Motivated by these observations, this article focuses on cross-domain object detection from VLRSIs to HRSIs.

B. Cross-Domain Object Detection Methods

1) Adversarial Learning-Based Methods: Adversarial learning-based methods [29] aim to learn domain-invariant representations [30], [31] by employing various domain discriminators [32]. Most existing approaches follow the domain adaptation (DA)-Faster region-based convolutional neural network (R-CNN) framework [33], [34], [35], [36], which introduces image-level and instance-level adaptation modules to mitigate domain shift [37], [38]. The image-level module primarily captures style and illumination variations [39], [40], whereas the instance-level module addresses differences in object appearance [41] and scale [42], [43]. To overcome the limitations of pixel-level adaptation (incomplete transformations) and feature-level adaptation (source bias), Kim et al. [44] propose domain diversification (DD) and multidomain-invariant representation learning (MRL).

While most prior methods emphasize image- and feature-level alignment, Yu et al. [46] observe that adversarial strategies mainly reduce style discrepancies but often neglect content shift, which is crucial for accurate object detection. To address this, they introduced style- and content-aware [45] unsupervised DA (SC-UDA) [46], which jointly reduces both style and content gaps. Moreover, Zhou et al. [47] propose multigranularity alignment DA (MGADA), which enhances domain-invariant learning by modeling dependencies across pixel-, instance-, and category-level representations.

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However, these adversarial learning-based methods primarily focus on reducing feature-level discrepancies such as appearance, illumination, and style shifts between RGB domains. They implicitly assume that both domains share similar modality structures and do not account for cross-modal discrepancies arising from spectral dimensionality differences. In the VLRSI–HRSI scenario, the target domain contains high-dimensional spectral signatures that are absent in the source domain, making simple feature-level alignment insufficient. Moreover, adversarial alignment may inadvertently suppress discriminative spectral cues critical for hyperspectral detection.

2) Self-Training-Based Methods: Self-training-based approaches, often regarded as a branch of semi-supervised learning, generally operate in two sequential phases [48]. Initially, a detector is trained on annotated data from the source domain [49]. In the subsequent phase, the unlabeled target-domain samples are processed by this detector to generate pseudo-annotations [50], which are then exploited to refine the model through self-supervised retraining [51]. Among these methods, teacher–student architectures have demonstrated strong potential in cross-domain object detection, yet their performance is often compromised by domain discrepancies, resulting in noisy or unreliable pseudo-annotations [52]. To mitigate this problem, Li et al. develop the adaptive teacher (AT) framework [53], which combines adversarial domain alignment with weak-to-strong augmentation to better bridge the gap between domains.

However, many existing studies overlook the uncertainty estimation of pseudo bounding boxes during the retraining phase. To address this issue, Chen et al. [54] propose the probabilistic teacher (PT) framework, which dynamically captures prediction uncertainty from the evolving teacher model and transfers it to the student through mutual regularization. Later, Zhao et al. [55] observe that teacher–student systems frequently generate inaccurate or incomplete pseudo-labels under severe domain discrepancy. They, thus, designed the masked retraining teacher–student (MRT) framework, which masks multiscale features of target-domain images and enables the student, equipped with an AE, to reconstruct the missing representations. This reconstruction mechanism allows the network to leverage contextual cues beyond the limited pseudo-annotations, enhancing overall learning robustness.

Later, Cao et al. [48] observe that existing self-training-based approaches heavily depend on pseudo-labels generated by teacher models, whose quality can be severely degraded by domain discrepancies. To address this limitation, they introduce contrastive mean teacher (CMT) variants—CMT-AT and CMT-PT—which integrate contrastive learning into existing self-training frameworks. By pulling together features of the same instance and pushing apart those of different instances, CMT effectively optimizes feature representations. Importantly, even in the presence of noisy pseudo-labels, contrastive learning enables the model to extract meaningful and beneficial learning signals.

Moreover, Chen et al. [56] highlight that most teacher–student detectors lack effective instance-level feature alignment in the decoder, as they typically adopt class-agnostic adaptation strategies that disregard interclass distinctions and limit detection precision. In addition, their batch-level alignment is often too localized, constraining generalization on the target domain. To overcome these drawbacks, they introduced DATR [56], a DEtection TRansformer (DETR)-based domain-adaptive detector [57] that performs multilevel instance alignment to achieve more stable cross-domain transfer.

Self-training-based approaches rely heavily on the quality of pseudo-labels generated from the source-trained detector. However, under significant cross-modal discrepancy between VLRSIs and HRSIs, pseudo-labels tend to be highly unreliable due to missing spectral information in the source domain. Without explicit modeling of spectral characteristics, these methods propagate noisy supervision into the target domain, which may lead to performance degradation rather than improvement. Furthermore, existing teacher–student frameworks are not designed to handle spectral uncertainty inherent in hyperspectral imagery.

In summary, although previous cross-domain object detection methods have demonstrated strong performance in conventional RGB-to-RGB adaptation scenarios, they are fundamentally built upon the assumption of modality consistency between source and target domains. In contrast, the VLRSI–HRSI setting involves not only appearance shift but also cross-modal discrepancy caused by the absence of high-dimensional spectral information in the source domain. Existing adversarial and self-training-based frameworks primarily address feature distribution alignment but do not explicitly model domain-shared spatial sparsity or spectral uncertainty. As a result, directly applying them to VLRSI–HRSI adaptation often leads to suboptimal alignment, unreliable pseudo-supervision, and degraded detection performance. These limitations motivate the development of a cross-domain framework that jointly considers residual-based spatial alignment and spectral uncertainty modeling.

Recent studies have explored DA techniques for hyperspectral image (HSI) analysis. For example, the two-branch attention adversarial DA (TAADA) network introduces a spectral–spatial feature extraction framework combined with adversarial learning to improve cross-domain HSI classification performance. By jointly modeling spectral and spatial information through a two-branch attention mechanism, TAADA effectively enhances the discriminative capability of learned features. These studies demonstrate the importance of exploiting spectral–spatial joint representations for cross-domain learning in hyperspectral imagery. Inspired by these observations, this work further investigates cross-domain learning in the context of hyperspectral object detection, where both spatial localization and spectral variability must be considered.

III. Methodology

A. Overview

1) Training Stage: In remote sensing image datasets, the proportion of objects is considerably smaller than that of the background. As a result, the reconstructed residuals tend to preserve richer object-related information and can, therefore, serve as domain-shared spatial representations between

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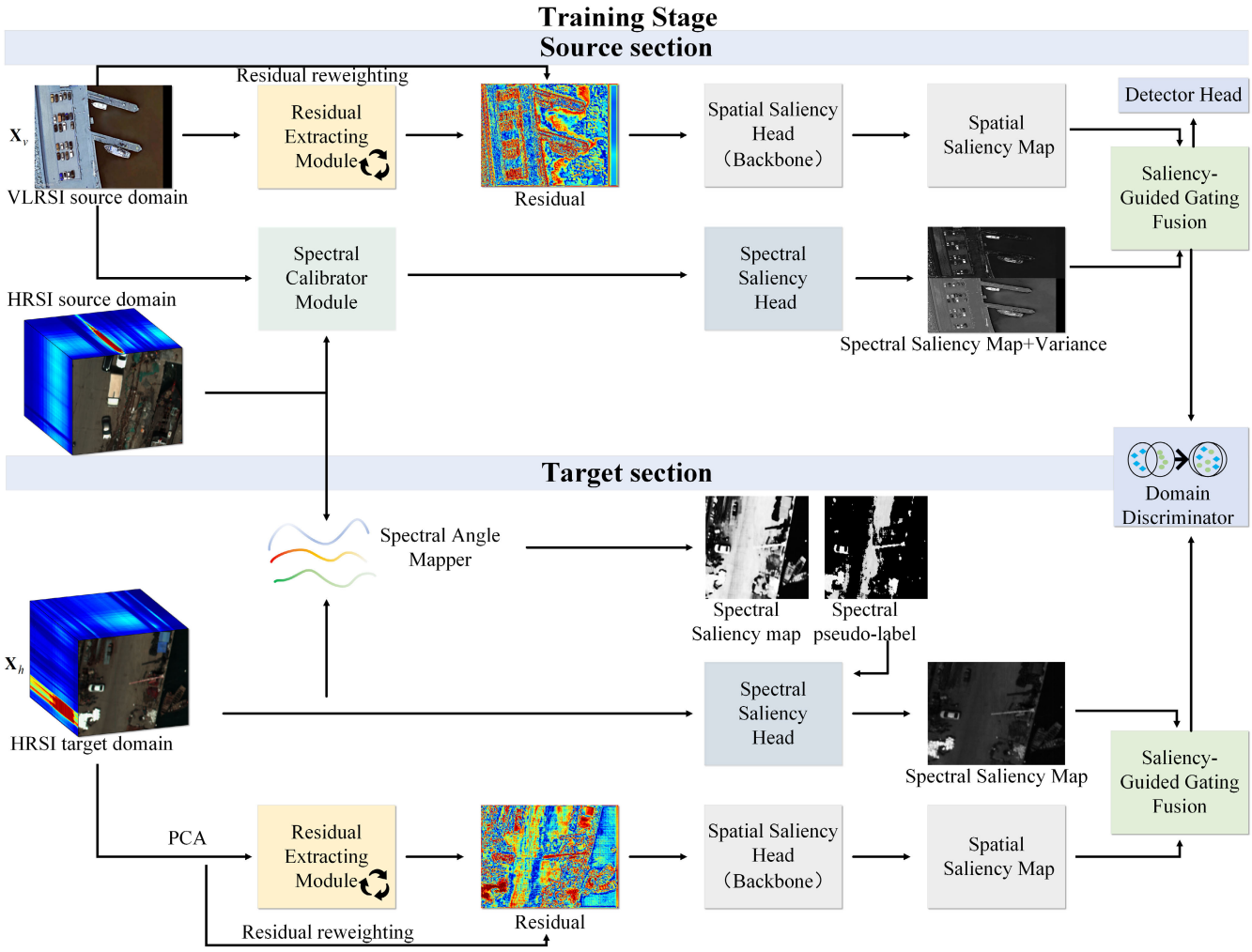


Fig. 2. Overall structure of our method in the training stage.

VLRSIs and HRSIs. Motivated by this observation, a residual extraction module is designed to explicitly extract the shared spatial cues across domains. The extracted residual features are reweighted with the original three-band image and then fed into the spatial saliency head to generate a spatial saliency map that highlights potential target regions.

To model the spectral features and their inherent uncertainty, an SSH is introduced. However, since spectral information is not available in the VLRSIs source domain, an SCM is designed for the VLRSIs source domain to generate spectral representations that are consistent with those in the HRSI target domain. The SCM predicts the coefficients of RBFs through a convolutional network and is constrained by a double softmin loss guided by the prior spectra [58] of one pre-collected HRSI. This mechanism enables objects with different semantic categories in the source domain to produce spectral curves similar to those of the target domain. The calibrated images are then processed by the SSH to generate spectral saliency maps corresponding to regions of interest. This head is trained using a combination of spectral heteroscedastic loss and a spectral uncertainty loss, which together transfer the spectral uncertainty learned from the labeled source to the unlabeled target domain.

Finally, the spatial and spectral saliency maps are fused through a saliency-guided gated fusion mechanism to obtain spatial–spectral fused features. These fused representations are further aligned through a domain discriminator based on gradient reversal layer (GRL) [32], and the detection head is trained exclusively on the labeled source domain since annotations are unavailable in the target domain. The overall training process is given in Fig. 2.

2) Inference Stage: After training, the model parameters of the REM, spatial saliency head, SCM, SSH, and detection head are obtained. During the inference stage, neither prior spectral curves nor source-domain VLRSIs are required, and only target-domain HRSIs are used as input. Spatially, the HRSIs are first reduced to three bands through principal component analysis (PCA) [59] and then fed into the pretrained REM. The extracted residual features are reweighted with the previous three-band images to serve as inputs for the spatial saliency head to generate the spatial saliency map. Spectrally, the pretrained SSH is directly applied to produce the spectral saliency map. The spatial and spectral saliency maps are then fused via a saliency-guided gating fusion mechanism to obtain spatial–spectral fused features, which are finally fed into the

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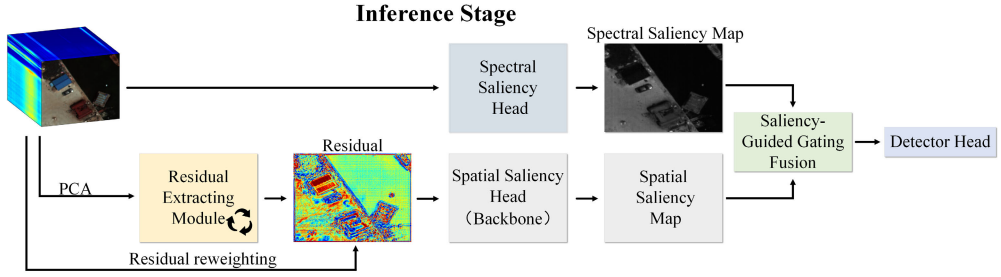


Fig. 3. Flowchart of the inference stage of the proposed method.

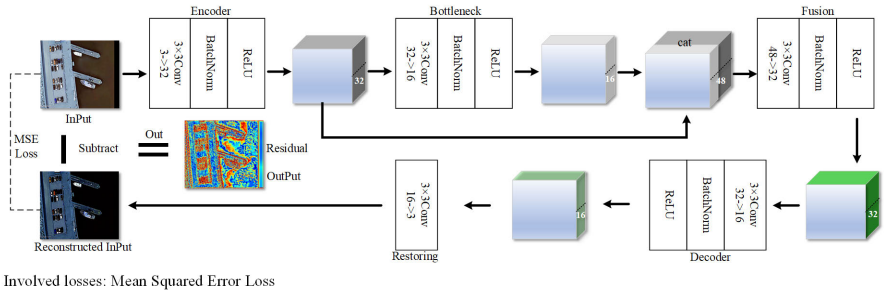


Fig. 4. Flowchart of the proposed residual extracting module.

detection head to accomplish object detection. The overall inference process is illustrated in Fig. 3.

B. Residual Extracting Module (REM)

To extract domain-shared spatial representations between the VLRSI, denoted as $\mathbf{X}_v \in \mathcal{R}^{H_1 \times W_1 \times 3}$, and HRSI, denoted as $\mathbf{X}_h \in \mathcal{R}^{H_2 \times W_2 \times C}$, we design the REM based on a self-reconstruction mechanism, as illustrated in Fig. 4. Here, W_1, H_1, W_2, H_2, and C denote the width and height of the source-domain VLRSI, the width and height of the target-domain HRSI, and the number of spectral bands, respectively.

Given the input image $\mathbf{X} \in \{\mathbf{X}_v, PCA(\mathbf{X}_h)\}$, the REM aims to decompose it into a reconstructed component $\dot{\mathbf{X}}$ and a residual component $\mathbf{R}$, where the residual emphasizes object-related spatial cues that are common across domains but difficult to reconstruct. Conventional residual extraction methods [such as directly computing $\mathbf{X}$-blur($\mathbf{X}$)] can only capture low-level pixel differences, whereas the proposed REM performs feature compression and reconstruction in the deep feature space, effectively achieving a semantic-level self-reconstruction difference.

The input image $\mathbf{X}$ is first processed by an encoder to extract local spatial structures

$$\mathbf{F}_{enc} = \sigma\left(\mathrm{BN}\left(\mathbf{W}_1 * \mathbf{X} + \mathbf{b}_1\right)\right) \tag{1}$$

where * denotes the convolution, and $\sigma(.)$ is the rectified linear unit (ReLU) function. BN(.) stands for batchnorm operation. This stage expands the channel dimension from 3 to 32, providing an over-complete spatial representation that captures richer edge, texture, and geometric cues while suppressing pixel-level noise.

The encoder output is further compressed through the bottleneck

$$\mathbf{F}_{bot} = \sigma\left(\mathrm{BN}\left(\mathbf{W}_2 * \mathbf{F}_{enc} + \mathbf{b}_2\right)\right). \tag{2}$$

This bottleneck compression reduces the feature dimension ($32 \rightarrow 16$) to form a compact latent representation for efficient reconstruction.

To recover the spatial details lost during compression, the bottleneck features are concatenated with the shallow encoder features and then passed through a fusion convolution

$$\mathbf{F}_{fus} = \sigma\left(\mathrm{BN}\left(\mathbf{W}_3 * [\mathbf{F}_{bot}, \mathbf{F}_{enc}] + \mathbf{b}_3\right)\right) \tag{3}$$

where [., .] denotes the concatenation along the channel dimension. This fusion step integrates shallow and deep features (48–32 channels) to maintain spatial continuity and recover fine structural details. The fused features are then decoded through two consecutive convolutional blocks

$$\begin{aligned} \mathbf{F}_{dec} &= \sigma\left(\mathrm{BN}\left(\mathbf{W}_4 * \mathbf{F}_{fus} + \mathbf{b}_4\right)\right) \\ \dot{\mathbf{X}} &= \mathbf{W}_5 * \mathbf{F}_{dec} + \mathbf{b}_5. \end{aligned} \tag{4}$$

The output $\dot{\mathbf{X}}$ represents the reconstructed input, optimized to minimize the mean-squared reconstruction loss

$$\mathcal{L}_{rec} = \left\|\mathbf{X} - \dot{\mathbf{X}}\right\|_F^2. \tag{5}$$

Finally, the residual $\mathbf{R}$ is obtained by subtracting the reconstructed image from the input

$$\mathbf{R} = \mathbf{X} - \dot{\mathbf{X}}. \tag{6}$$

The residual highlights the regions that are difficult to reconstruct (object regions), which usually correspond to domain-shared spatial structures. These residuals are then used for reweighting of the three-band image, and their mathematical formula can be expressed as

$$\tilde{\mathbf{X}} = \mathbf{X} + \frac{std\left(\mathbf{X}\right)}{std\left(\mathbf{R}\right)}\mathbf{R} \tag{7}$$

where $\tilde{\mathbf{X}} \in \{\tilde{\mathbf{X}}_v, \tilde{\mathbf{X}}_h\}$ denotes the residual reweighted variable, which is used as the input of the backbone (spatial saliency head).

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C. Spectral Saliency Head

During training, the processing procedure and the generated losses of the spectral branch differ between domains. In the source domain, the SCM calibrates the spectra of VLRSIs based on the prior spectra of the additional HRSI, and the calibrated results are then fed into the SSH. Guided by the ground truth (GT), a spectral heteroscedastic loss is applied to learn the spectral uncertainty of the source domain. In contrast, in the HRSI target domain, the SSH directly produces the spectral saliency map, which is combined with the SAM [17] metric to generate a saliency map. High-confidence pixels are extracted as pseudo-labels, and a binary cross-entropy loss is employed to learn the spectral uncertainty of the HRSI target domain. Therefore, we separately describe the processing procedures for the source and target domains.

1) Source-Domain-Related: In the spectral branch processing of the source-domain VLRSI, it is first necessary to enable the VLRSI to generate reasonable spectral features. To this end, we propose the SCM. The SCM takes the source-domain VLRSI and the prior spectral curves $\mathbf{S} \in \{\mathbf{S}^{(cl)} \in \mathbb{R}^{C\times Cl\times P}; \mathbf{S}^{(bg)} \in \mathbb{R}^{C\times 1}\}$ as inputs, where the Cl denotes the number of categories, and P represents the number of spectral diversities, meaning that each category contains several representative spectral curves. It should be noted that the prior spectral curve comes from a precollected HRSI. To generate continuous and smooth spectral curves, we employ a CNN to predict the coefficients of the RBFs. Accordingly, we first define the construction process of the RBF basis. Let the wavelength range $\mathbf{wr}$ be

$$\mathbf{wr} = [wr_1, wr_2, \ldots, wr_C]^T, wr_i \in [wl_{mi}, wl_{ma}] \tag{8}$$

where wl_{mi} and wl_{ma} denote the minimum and maximum wavelengths of the target-domain HRSI, respectively. We uniformly choose K RBF centers, within this range, where K is fixed to 48 in our method. Consequently, each RBF center can be defined as

$$\boldsymbol{\mu} = \left[\mu_1, \mu_2, \ldots, \mu_K\right]^T, \mu_j = wl_{\mathrm{mi}} + \frac{(j-1)}{K-1}\left(wl_{ma} - wl_{\mathrm{mi}}\right). \tag{9}$$

Each RBF basis function is defined as

$$\phi_j(wr_i) = \exp\left(-\frac{1}{2}\left(\frac{wr_i - \mu_j}{\sigma}\right)^2\right) \tag{10}$$

where σ controls the width of the basis and is set to $\sigma = (wr_{ma} - wr_{mi})/K \times \epsilon$. After normalization across the wavelength dimension

$$B_{ij} = \frac{\phi_j(wr_i)}{\sum_{i=1}^{C}\phi_j(wr_i) + \varepsilon}. \tag{11}$$

Therefore, we obtain the RBF basis matrix $\mathbf{B} = [B_{ij}] \in \mathbb{R}^{C\times K}$.

For each spatial position (h, w), the network predicts K nonnegative coefficients using a 1×1 convolution followed by a Softplus activation:

$$A_{b,j,h,w} = Softplus\left((\mathbf{W}_a * \mathbf{X}_v)_{b,j,h,w} + b_j\right) \tag{12}$$

resulting in the coefficient tensor $\mathbf{A} \in \mathbb{R}^{B\times K\times H\times W}$. Accordingly, the VLRSI is constructed as the linear combination of the RBF bases weighted by the coefficients

$$\hat{\mathbf{X}}_v = \sum_{j=1}^{K} B_{cj} A_{b,j,h,w}. \tag{13}$$

The entire reconstruction process can be compactly expressed as

$$\hat{\mathbf{X}}_v = \mathbf{B}Softplus\left(\mathbf{W}_a * \mathbf{X}_v + \mathbf{b}_a\right). \tag{14}$$

This design allows the SCM to approximate smooth, physically consistent spectral curves by learning only a small number of spectral bases.

To constrain the reconstructed spectra to be consistent with the category-specific spectral priors, we design a differentiable spectral matching loss based on a two-level soft-min aggregation of the SAM distance. Given the predicted hyperspectral image $\hat{\mathbf{X}}_v \in \mathbb{R}^{H_1\times W_1\times C}$ from VLRSI $\mathbf{X}_v$, the spectral matching loss $\mathcal{L}_{SM}$ for all object categories and background regions is formulated as

$$\varphi_{cl}(Bo) \coloneqq \underset{(i,j)\in\mathcal{P}(Bo)}{\mathrm{softmin}}\ \underset{1\le k\le P}{\mathrm{softmin}}\ \arccos\left(\hat{\mathbf{X}}_v^{(i,j)},\ \mathbf{S}_k^{(cl)}\right) \tag{15a}$$

$$\psi \coloneqq \underset{(i,j)}{\mathrm{softmin}}\ \arccos\left(\hat{\mathbf{X}}_v^{(i,j)},\ \mathbf{S}^{(\mathrm{bg})}\right) \tag{15b}$$

$$\mathcal{L}_{sm} = \sum_{cl\in Cl}\frac{1}{|Bo_{cl}|}\sum_{Bo\in Bo_{cl}}\varphi_{cl}(Bo) + \psi. \tag{15c}$$

The $\mathcal{P}(Bo)$ refers to extracting all the pixels from a given bounding box Bo. Through the constraint of the above losses, the spectra generated by the SCM correspond to the prior spectra of different semantic categories in the target domain, while the prior spectra of nonobject regions are encouraged to approach the background spectra.

To generate the spectral saliency map, the generated $\hat{\mathbf{X}}_v$ is fed into the SSH, where, for simplicity, the corresponding mathematical expression is denoted as SSH$(\cdot)$. Through computation, the saliency map and variance are obtained

$$ssm_v, var_v = \mathrm{SSH}\left(\hat{\mathbf{X}}_v\right). \tag{16}$$

For a clearer understanding of the proposed module, its workflow is visualized in Fig. 5. The spectral heteroscedasticity loss is introduced to model input-dependent spectral uncertainty in the source domain. In HSI, the spectral responses of pixels from the same semantic category may still fluctuate due to material variation, environmental conditions, and imaging noise. Therefore, instead of assuming a uniform confidence level for all pixels, we allow the SSH to predict a data-dependent uncertainty for each spectral response. The proposed loss adaptively downweights the binary cross-entropy term for uncertain pixels, while a regularization term constrains the predicted uncertainty to avoid trivial solutions. In this way, the model learns more robust spectral saliency representations under intraclass spectral variability

$$\mathcal{L}_{sh} = \frac{1}{|\Omega|}\sum_{(i,j)\in\Omega}\left(e^{-var_v^{(i,j)}}\,\mathrm{BCEWithLogits}\left(ssm_v{}^{(i,j)},\ y^{(i,j)}\right) + var_v^{(i,j)}\right) \tag{17}$$

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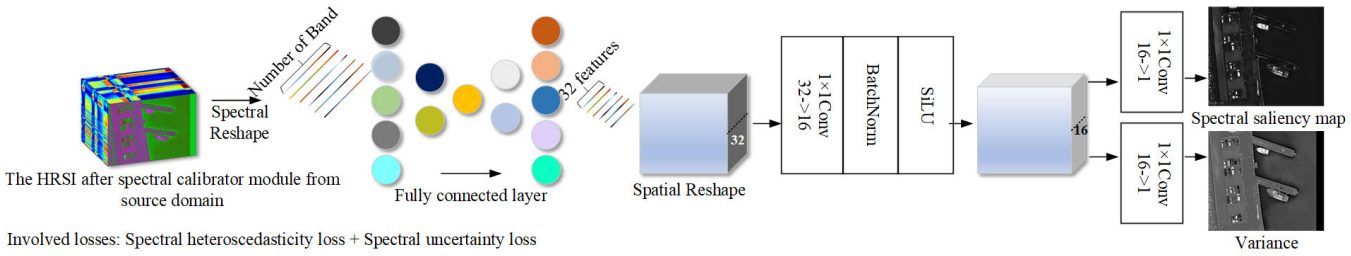


Fig. 5. Flowchart of the SSH.

where $y^{(i,j)} \in 0, 1$ denotes the binary target indicating whether pixel (i, j) belongs to the foreground spectrum or the background, which is generated by the GT bounding box, and Ω is the spatial set of all valid pixels. The first term in (17) adaptively reweights the binary cross-entropy according to the predicted uncertainty, reducing the contribution of pixels with higher spectral variance, while the second term $var^{(i,j)}$ acts as a regularizer that penalizes excessive variance, encouraging the model to reduce uncertainty in confident regions and preventing trivial inflation of variance.

This design enables the SSH to learn input-dependent (heteroscedastic) spectral uncertainties, improving robustness against intraclass spectral variability and illumination-induced spectral ambiguity.

2) Target-Domain-Related: In the absence of GT in the target domain, we generate pseudo-spectral saliency maps based on the SAM as pseudo-labels. High-confidence pixels are selected using a top-p% plus soft-weighting mechanism for supervision, and a weighted binary cross-entropy loss is employed to stably learn the spectral uncertainty of the target domain.

First, the generation process of SAM can be formulated as follows:

$$S_{\mathrm{SAM}}^{(i,j)} = 1 - \frac{1}{\pi} \min_{1 \le k \le Cl \times P} \arccos\left(\mathbf{X}_h, \mathbf{S}_k^{(cl)}\right) \tag{18}$$

where $S_{\mathrm{SAM}}^{(i,j)}$ denotes the spectral saliency value at pixel (i, j).

To generate pseudo-labels, we introduce a confidence-guided pixel weighting mechanism. The core idea of this mechanism is to select only the top-p% most confident pixels in the spectral saliency map as a "hard mask," and then combine them with a soft exponential weighting to form the final weight map

$$we^{(i,j)} = \mathbf{1}\left[S_{\mathrm{SAM}}^{(i,j)} \ge q_p\right] \cdot \exp\left(-\frac{1 - S_{\mathrm{SAM}}^{(i,j)}}{\tau}\right) \tag{19}$$

where q_p represents the $(1 - p)$ quantile of S_{SAM} within the image (i.e., the threshold corresponding to the top-p% most confident pixels). τ is a temperature parameter controlling the softness of the exponential weighting, and $\mathbf{1}[.]$ denotes the indicator function that equals 1 when the condition inside is true and 0 otherwise. It is noted that in our experiments, the parameters p and τ are by default set to 0.2 and 0.4, respectively.

The parameter $p = 0.2$ is introduced to select high-confidence pixels from the SAM response map. These pixels are most likely to correspond to object regions, and only high-confidence labels provide the most valuable supervision for model learning during training. Once the model has been trained, the detection process becomes independent of this confidence threshold. When hyperspectral scenes with high object density are input, the model can automatically highlight pixels whose spectra are similar to the object spectra, regardless of the number of object pixels present in the scene.

It should be noted that the pseudo-labels used in this work differ from those in conventional self-training frameworks. In typical teacher–student methods, pseudo-labels are generated from the predictions of a detector and are iteratively updated during training, resulting in confidence distributions that evolve across epochs. In contrast, the pseudo-labels in this study are derived from the SAM, which measures the spectral similarity between each pixel and the prior spectral signatures. Specifically, a spectral saliency map is first computed using SAM, and the top-p% pixels with the highest saliency values are selected as pseudo-labels. These pixels are further assigned confidence weights through an exponential weighting function based on their SAM scores. Since the pseudo-labels are determined directly from spectral similarity rather than model predictions, they remain fixed during training and do not exhibit epochwise confidence evolution.

To account for the inherent spectral variability in the target-domain HRSI, we introduce a spectral uncertainty loss. In practical hyperspectral imaging scenarios, spectral responses are affected by multiple factors, including illumination variation, atmospheric interference, sensor noise, and material heterogeneity. These factors induce uncertainty and dispersion in spectral signatures, especially in cross-domain settings where acquisition conditions differ.

The spectral uncertainty loss is, therefore, designed to model this input-dependent uncertainty and to improve robustness in the spectral space. By assigning confidence-aware weights to pseudo-labeled pixels, the loss stabilizes the learning process under noisy or ambiguous spectral responses and prevents unreliable pseudo-labels from dominating the optimization.

Accordingly, the spectral uncertainty loss for the target-domain HRSI is formulated as follows:

$$ssm_h = \mathrm{SSH}\left(\mathbf{X}_h\right) \tag{20}$$

$$\mathcal{L}_{\mathrm{spec}} = \frac{\sum_{i,j} we^{(i,j)} \cdot \mathrm{BCEWithLogits}\left(ssm_h^{(i,j)}, S_{\mathrm{SAM}}^{(i,j)}\right)}{\sum_{i,j} we^{(i,j)} + \varepsilon}. \tag{21}$$

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D. Saliency-Guided Gating Fusion

In the spatial branch, when the features extracted by the REM are passed through the backbone, a spatial saliency map fp [feature pyramid network (FPN) features)] is obtained by $fp = Backbone(\tilde{\mathbf{X}})$; in the spectral branch, the SSH generates the spectral saliency map ssm.

To achieve effective fusion between spatial and spectral features, we introduce a saliency-guided gating fusion mechanism. The basic principle is to use the spectral or spatial saliency maps to enhance the feature maps at each level of the FPN. Under the constraints of nonnegativity and amplitude limitation, salient regions are proportionally amplified. The corresponding process can be formulated as follows:

$$\hat{fp}_l = fp_l \odot (1 + \alpha_l . ssm). \tag{22}$$

Here, α_l denotes the fusion parameter set for the lth layer, which is empirically assigned as 1.0, 0.5, 0.25, and 0.125 for layers 2–5, respectively.

The rationale behind this design is to assign larger weights to features at larger spatial scales and smaller weights to those at smaller spatial scales. This is because applying a smaller weight to the spectral features at smaller spatial scales allows the model to tolerate certain prediction errors. In contrast, if larger weights were assigned to the spectral features at smaller spatial scales, the model would require these spectral representations to be highly accurate. However, since the proposed method introduces only a limited number of single hyperspectral images during training, the available prior spectral information is relatively limited. Therefore, this weighting strategy allows a certain degree of error in the spectral representations, enabling the model to compensate through complementary spatial features for effective object detection.

Finally, in the source domain, the enhanced features are fed into both the detection head and the domain classifier, while in the target domain, the enhanced features are fed only into the domain classifier. The structure of the domain classifier is shown in Fig. 6, which ensures that during the backpropagation process, no domain is favored, making it difficult to distinguish whether the features originate from the source domain or the target domain.

E. Overall Loss of the Proposed Method

The overall loss consists of three parts: 1) the detection head loss $\mathcal{L}_{\text{RPN_RoI}}$, spectral matching loss $\mathcal{L}_{\text{sm}}$, and spectral heteroscedasticity loss $\mathcal{L}_{\text{sh}}$, which are computed only on the source domain; 2) the spectral uncertainty loss $\mathcal{L}_{\text{spec}}$, which is computed only on the target domain; and 3) the reconstruction loss $\mathcal{L}_{\text{rec}}$ and domain discriminator loss $\mathcal{L}_{\text{adv}}$, which involve both source and target domains. Accordingly, the overall loss can be formulated as follows:

$$\begin{aligned}\mathcal{L} = \mathcal{L}_{\text{RPN_RoI}} + \kappa\mathcal{L}_{\text{rec}} \\ + \beta\mathcal{L}_{\text{sm}} + \gamma\mathcal{L}_{\text{sh}} + \lambda\mathcal{L}_{\text{spec}} + \zeta\mathcal{L}_{\text{adv}}.\end{aligned} \tag{23}$$

Here, κ, β, γ, λ, and ζ denote the hyperparameters, which are uniformly set to 0.1, 0.001, 0.001, 0.001, and 0.0001, respectively, during the experiments.

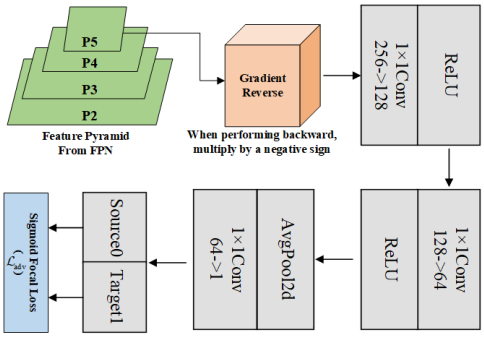


Fig. 6. Schematic of domain classifier.

IV. Experiments and Analysis

In this section, we conduct comparative experiments under three experimental settings to verify the effectiveness of the proposed method. In addition, ablation studies are carried out to validate the effectiveness of the proposed modules.

A. Dataset Description

1) Target-Domain Dataset:

a) Lwp2.0: Due to the high cost of acquiring hyperspectral data (HSI), large-scale remote sensing hyperspectral datasets involving semantic information are scarce. To date, the only known large-scale HRSI datasets involving semantic information are M2SODAI [60] and LWP [61]. Among them, M2SODAI contains the ship and marine debris categories; however, marine debris is extremely scarce in the VLRSI datasets. The LWP dataset, proposed by our team, originally included only the ship category. To construct a dataset with multiple semantic categories, we further refined and extended it to create LWP2.0, which serves as the target domain in our experiments. The detailed parameters are provided in Table I.

The LWP2.0 dataset was obtained using the Nano-Hyperspec sensor. This sensor captures HSIs across 270 spectral bands, covering the wavelength range from 400 to 1000 nm. The dataset is composed of 627 HSIs, each having a spatial dimension of 224×224 pixels. These images possess a high spatial resolution, which ranges from 0.1 to 0.2 m, along with a fine spectral resolution of 2.2 nm.

b) M2SODAI: M2SODAI is a multimodal dataset that includes both RGB images and HSIs. In this study, only the HSI component of the dataset is utilized. The HSI data are acquired by the AsiaFENIX sensor and contain 127 spectral bands covering the wavelength range from 400 to 1000 nm. The original dataset contains two object categories: floating matter and ships. To maintain semantic consistency with the categories used in the source-domain datasets, only the ship category is retained in this study. The processed M2SODAI dataset used in this work consists of 1132 individual HSIs, each with a spatial dimension of 224×224 pixels. These images have a spatial resolution of 0.7 m and a spectral resolution of 4.5 nm. For comprehensive evaluation, the dataset is divided into two subsets: a training set composed of 1007 HSIs containing 4174 ship instances and a test set comprising 125 HSIs with 511 ship instances.

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TABLE I
DETAILED STATISTICS OF THE TARGET-DOMAIN DATASET

LWP2.0:270 bands 400 nm-1000 nm							
		small($< 32^2$)	medium($32^2 - 96^2$)	large($> 96^2$)	total	Number of HSIs	Number of instances
training	ship	165	436	120	721	492	886
	car	61	23	0	84		
	long vehicle	38	43	0	81		
inference	ship	70	144	26	240	135	279
	car	18	1	0	19		
	long vehicle	12	8	0	20		
M2SODAI:270 bands 400 nm-1000 nm							
		small($< 32^2$)	medium($32^2 - 96^2$)	large($> 96^2$)	total	Number of HSIs	Number of instances
training	ship	3980	193	1	4174	1007	4174
inference	ship	496	15	0	511	125	511

2) Source-Domain Dataset: The choice of the satellite imagery multivehicle dataset (SIMD) and multivertical-perspective object detection (MVPOD) dataset is determined by the semantic categories present in the target-domain dataset, LWP2.0. LWP2.0 includes three categories: ship, car, and long vehicle. By analyzing different datasets and consolidating their categories, we found that both the SIMD and MVPOD datasets contain these three categories. Therefore, we selected the SIMD [62] and MVPOD [63] dataset as the source domain, as they align with the categories in LWP2.0.

a) SIMD: SIMD was released by Haroon et al. [62], with instance annotations provided in horizontal bounding box format. The dataset was collected from Google Earth and contains 15 categories, including cars, trucks, buses, long vehicles, various types of aircraft, and ships. It consists of 5000 images with a total of 45 096 annotated instances, and the spatial resolutions of the images are 1024×768 and 1280×720. Since SIMD is used only as a source-domain dataset, we consider only its training set. The original dataset statistics are shown in Table II. During use, to align the semantic categories with those of the target domain, we select boat, car, and long vehicle as the main categories. It is noted that the bus, truck, and van classes share similar semantics with long vehicle; therefore, they are merged into the long-vehicle category. The resulting dataset configuration is summarized in Table III.

Additionally, to test the cross-domain robustness under other hyperspectral sensors, we filtered out other categories from the SIMD, leaving only the "ship" class, in order to perform category matching under the SIMD to M2SODAI cross-domain setup.

b) MVPOD: MVPOD was released by Jin et al. [63], with instance annotations provided in horizontal bounding box format. The dataset was collected from three sources: a subset of the HRRSD dataset [64], Da-Jiang Innovations (DJIs) drone imagery, and Google Earth. It contains eight categories—airplane, bridge, bus, car, cargo ship, carrier, truck, and warship—comprising 10 470 images with a total of 15 467 annotated instances. The spatial dimensions of the images vary widely, with both width and height ranging from 200 to 11 000 pixels. Since MVPOD is used only as a source-domain dataset, we consider only its training set. The original category statistics are presented in Table IV. To align the semantic categories with those of the target domain, we merge cargo ship, carrier, and warship into the ship category, and merge bus and truck into the long-vehicle category, forming a dataset with semantic categories consistent with the target domain, as shown in Table V.

TABLE II
STATISTICS OF THE ORIGINAL SIMD

	No. instances	No. instances	No. images
airliner	788	36677	4000
boat	7140		
bus	1586		
car	16515		
chartered aircraft	547		
fighter aircraft	36		
helicopter	51		
long vehicle	1321		
propeller aircraft	153		
pushback truck	160		
stair truck	340		
trainer aircraft	548		
truck	2207		
van	4650		
others	635		

TABLE III
STATISTICS OF THE FILTERED SIMD

	No.instances	No. instances	No. images
ship	7140	33419	3228
car	16515		
long vehicle	9764		

B. Experimental Setup

Note that for the convenience of writing, we set the cross-domain experiment from the SIMD to the LWP2.0 dataset to be abbreviated as SIMD2LWP2.0, and the corresponding cross-domain experiment from the MVPOD dataset to the LWP2.0 dataset to be abbreviated as MVPOD2LWP2.0. Additionally, the cross-domain experiment from the SIMD to the M2SODAI dataset is abbreviated as SIMD2M2SODAI.

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TABLE IV
STATISTICS OF THE ORIGINAL MVPOD DATASET

	No. instances	No. instances	No. images
airplane	1692	12396	8376
bridge	1454		
bus	911		
car	3008		
cargoship	1988		
carrier	560		
truck	1224		
warship	1559		

TABLE V
STATISTICS OF THE FILTERED MVPOD DATASET

	No.instances	No. instances	No. images
ship	4107	9250	6216
car	3008		
long vehicle	2135		

1) Comparison Methods: Since there are currently no methods specifically designed for cross-domain detection from VLRSI to HRSI, nor for HRSI-to-HRSI cross-domain object detection, this article adopts several general RGB-based cross-domain object detection methods for comparison. Seven representative methods, denoted as AT, CMT-AT, PT, CMT-PT, MRT, MGADA, and DATR, are selected as baselines. These methods have achieved remarkable success in conventional RGB cross-domain detection tasks. For fair comparison, all methods are trained with the same number of iterations as the proposed approach, i.e., 15 000 iterations, and other hyperparameters are set according to the recommended configurations in their respective original articles. It is worth noting that, to make these methods applicable to HRSI data, the same experimental setting as ours is adopted, where PCA is used to reduce the dimensionality of HRSI images.

During training, several data augmentations are applied to the RGB images, including horizontal flipping, random rotation, ResizeShortestEdge, Gaussian blur, brightness perturbation, and random contrast. In contrast, no augmentation is applied to the hyperspectral images in order to preserve their spectral reflection characteristics. The backbone network is ResNet-101, initialized with the official pretrained weights provided by Detectron2. The model is optimized using stochastic gradient descent (SGD) with a momentum of 0.9. A linear warmup learning rate schedule is adopted, where the learning rate increases linearly from 5×10^{-6} to the base learning rate of 5×10^{-3} during the first 1000 iterations, and remains constant thereafter. All models are trained for 15 000 iterations with a batch size of 4, where each batch contains four RGB images and four hyperspectral images. It should be noted that the REM is trained jointly with the entire network in an end-to-end manner, rather than being trained separately.

2) Evaluation Metrics: The performance of the proposed method is evaluated both qualitatively and quantitatively. Specifically, our experiments include detection maps, mean average precision (mAP_{50}, mAP_{small}, mAP_{medium}, mAP_{large}), and heatmaps generated by different modules. The detection maps provide an intuitive qualitative assessment of the object

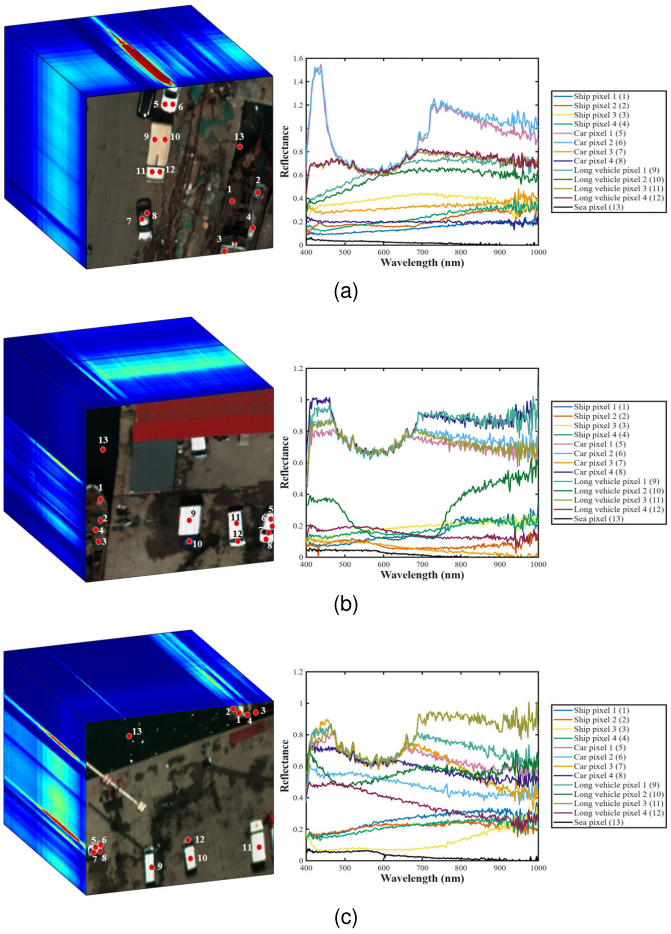


Fig. 7. Auxiliary spectral curves used in the involved LWP2.0 experiment (spectral prior). In the comparison experiments, the spectral prior from Scene 1 was used. The sensitivity analysis of spectral priors involved prior spectral curves from three different scenes. (a) Scene 1. (b) Scene 2. (c) Scene 3.

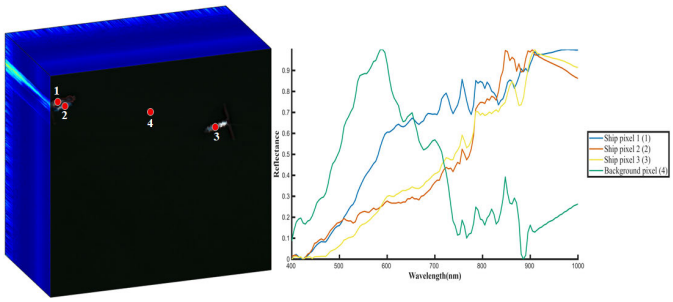


Fig. 8. Auxiliary spectral curves used in the involved M2SODAI experiment (spectral prior).

detection results. All quantitative metrics are computed using the pycocotools [65] library. In addition, the heatmaps are employed to qualitatively illustrate the feature representation capabilities of different modules.

All experiments are conducted on an Ubuntu 20.04 platform equipped with an Intel Core i9-9820X CPU, 64-GB RAM, and four NVIDIA GeForce RTX 2080 Ti GPUs (11-GB VRAM each). Both the baseline methods and the proposed approach are implemented using the PyTorch framework. During the operation of the SCM and the generation of SAM maps, prior spectral curves are required. To this end, we first collect an HRSI captured by the same Nano-Hyperspec sensor. From

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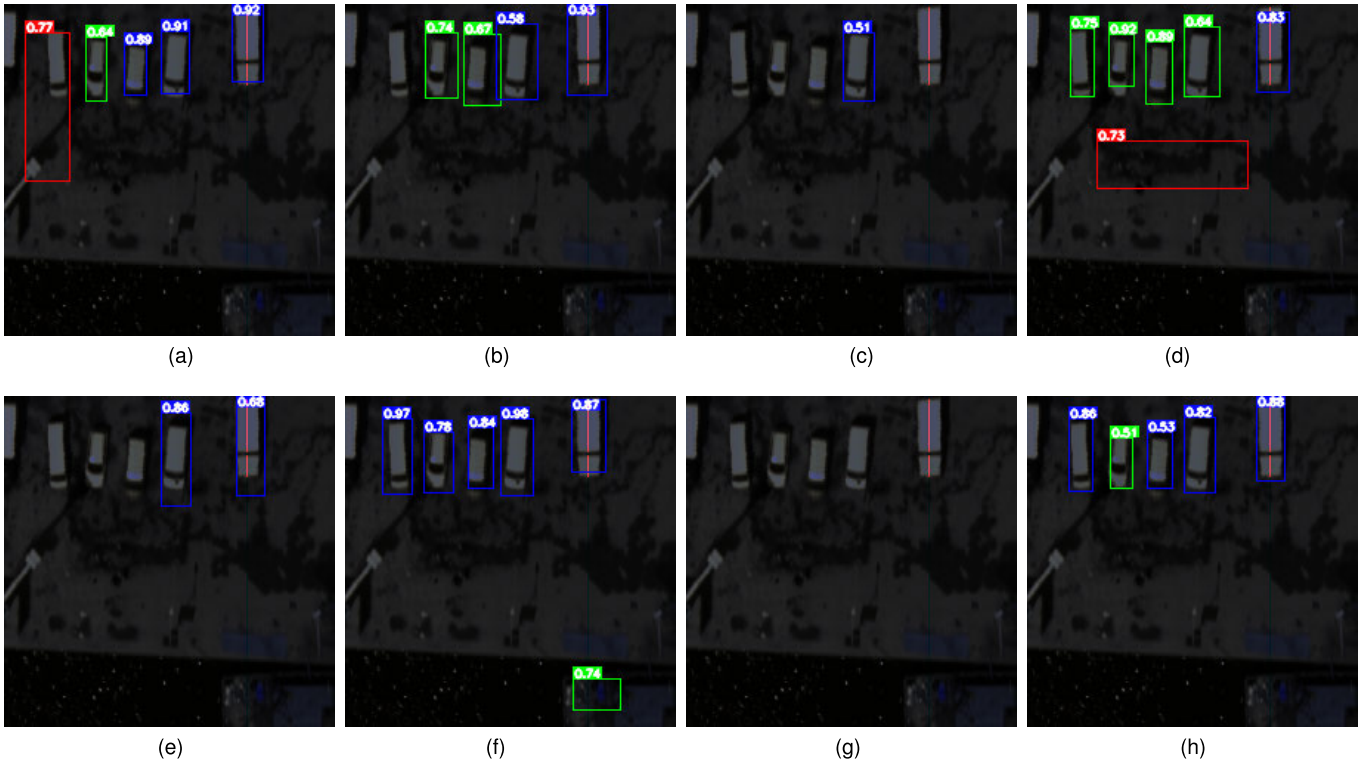


Fig. 9. Qualitative detection maps of different methods on SIMD2LWP2.0, while blue, green, and red boxes denote long vehicles, cars, and ships, respectively. (a) AT. (b) CMT-AT. (c) PT. (d) CMT-PT. (e) MRT. (f) MGADA. (g) DATR. (h) Ours.

some hyperspectral image, many spectral curves are selected for our experiment, as illustrated in Figs. 7 and 8.

C. Comparison Results

First, we conduct a qualitative experimental analysis of the SIMD2LWP2.0 experiment, as shown in Fig. 9.

As illustrated in Fig. 9, the qualitative results reveal three key issues observed in existing methods—pseudo-label unreliability, feature misalignment, and limited cross-domain generalization.

Methods based on self-training, such as AT and PT, struggle to generate reliable pseudo-labels under the VLRSI2HRSI scenario due to the large domain gap. AT mistakenly classifies the first long vehicle as a ship because the pseudo-labels generated from the source domain are highly uncertain. PT, though designed to model pseudo-label uncertainty, still relies on its correctness; consequently, only one long vehicle is detected with a low confidence score.

Contrastive learning-enhanced variants (CMT-AT and CMT-PT) attempt to mitigate domain discrepancy by aligning instance features. CMT-AT reduces false detections of long vehicles but still fails to detect the correct object. Other DA strategies show partial improvements but still exhibit inherent limitations. MRT employs selective retraining and masked autoencoding, which helps the model escape from suboptimal pseudo-label traps but struggles to align spectral–spatial domains, resulting in missed detections. MGADA introduces multigranularity alignment, which effectively reduces missed detections but misclassifies some objects (e.g., a car as a long vehicle) due to the lack of spectral awareness. DATR, built upon a transformer with one-to-one matching, generates conservative confidence scores; although it achieves high mAP, its visualized detections are incomplete.

In contrast, our method correctly detects and classifies both car and long-vehicle categories. This improvement mainly arises from the joint modeling of spatial and spectral features, which enhances both localization and classification robustness under large domain shifts.

Second, we conduct a qualitative experimental analysis of the MVPOD2LWP2.0 experiment, as shown in Fig. 10. This scenario further highlights the difficulty of aligning visible-light and hyperspectral domains, particularly for ship detection.

AT detects all ship targets but merges nearby floating nonobjects due to the absence of spectral constraints, causing bounding boxes to encompass multiple objects. CMT-AT, instead of improving localization, aggravates the adhesion of adjacent ships, likely because noisy pseudo-labels distort feature alignment during contrastive training. PT completely fails under this setting since unreliable pseudo-labels dominate training. CMT-PT alleviates this issue by emphasizing instance-level contrast rather than localization, leading to improved ship recognition accuracy.

MRT benefits from its masked autoencoder (MAE)-based retraining scheme, yet its learning remains confined to spatial context within the VLRSI modality. The absence of spectral alignment prevents it from bridging the cross-modal discrepancy, resulting in missed detections. In MGADA, multilevel feature fusion enables the method to effectively locate the location of objects, but it does not consider spectral features and cannot achieve effective truncation between nonobjects and objects.

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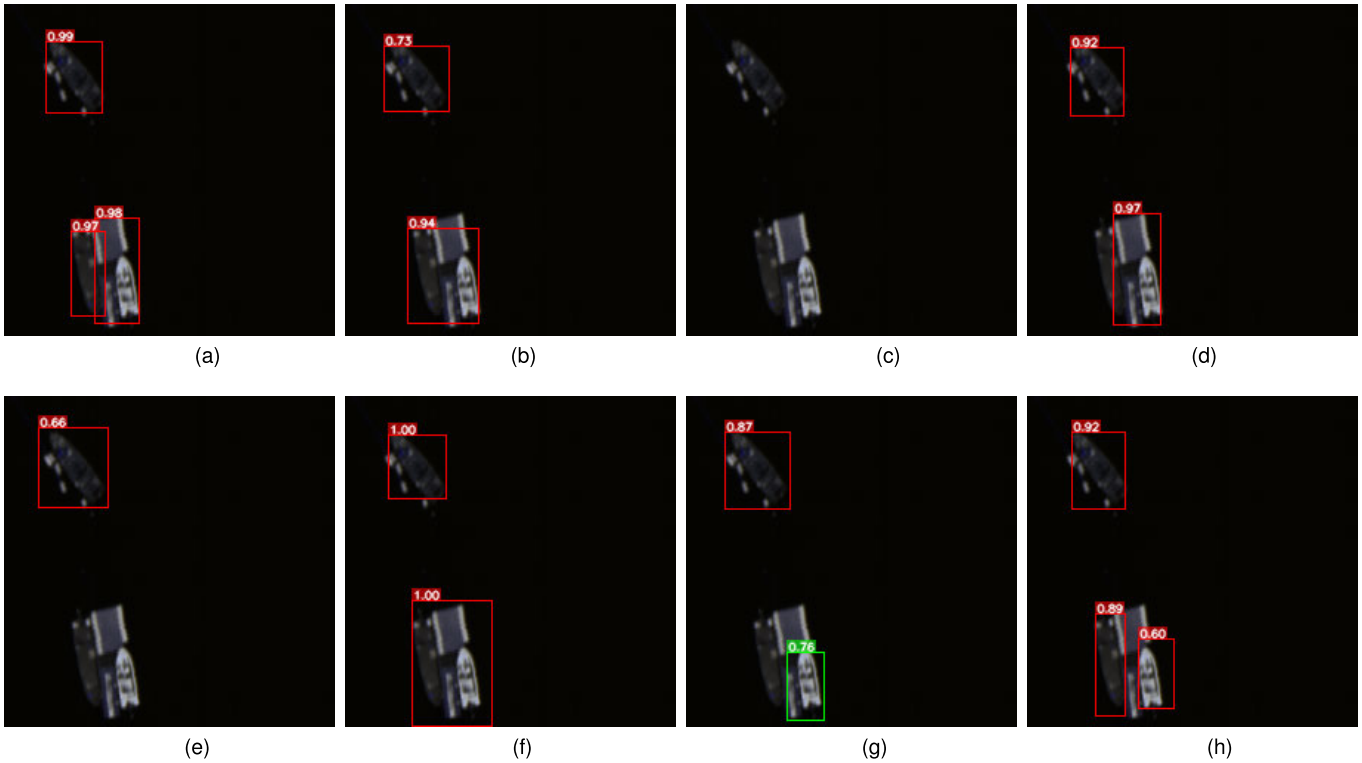


Fig. 10. Qualitative detection maps of different methods on MVPOD2LWP2.0, while blue, green, and red boxes denote long vehicles, cars, and ships, respectively. (a) AT. (b) CMT-AT. (c) PT. (d) CMT-PT. (e) MRT. (f) MGADA. (g) DATR. (h) Ours.

In the DATR method, the classwise prototype alignment (CPA) only aligns features in the visual space without considering spectral differences, resulting in cross-modal feature mismatches that cause ship targets to be misclassified as car or missed.

Our approach successfully detects all ship targets while avoiding nonobject inclusion, validating the effectiveness of spectral–spatial joint alignment in addressing both localization and classification inconsistencies.

Accordingly, the corresponding quantitative experimental results are recorded in Table VI. It can be seen that our method obtains the state of the art (SOTA) results. Except for the long-vehicle class of MVPOD2LWP2.0, our method obtains the best results in the two cross-domain settings.

In the SIMD2M2SODAI experimental setting, the object size distributions of the two datasets are highly inconsistent. Specifically, ships in the SIMD are generally large, whereas ships in the M2SODAI dataset are predominantly small. This substantial discrepancy makes it difficult for spatial features learned from RGB images to transfer effectively across domains. Most existing cross-domain object detection methods mainly focus on spatial feature alignment; however, such strategies become less effective under severe object-scale discrepancies. By introducing complementary spectral information, the proposed method alleviates the limitations of purely spatial feature transfer and achieves the best detection performance. Both the quantitative results (Table VI) and qualitative visualizations (Fig. 11) demonstrate the effectiveness of the proposed approach. In particular, as shown in Table I, small objects constitute the majority of targets in the M2SODAI dataset. Although some conventional cross-domain methods achieve relatively better performance on medium-sized objects, their overall mAP_{50} remains lower than that of the proposed method. These results further demonstrate the superiority and robustness of the proposed approach across different hyperspectral datasets.

To evaluate the sensitivity of the proposed method to the prior spectra, two additional hyperspectral images acquired at different times and from different regions are selected, from which new prior spectral curves are extracted. The experiments are conducted under the MVPOD2LWP2.0 cross-domain setting. The selected spectral curves are illustrated in Fig. 7, and the corresponding experimental results for the three scenes are reported in Table VII. The results show that replacing the prior spectral curves leads to only a slight decrease in performance, with fluctuations remaining within a small range. This indicates that the proposed method is not highly sensitive to the selection of prior spectra.

Furthermore, to further evaluate the sensitivity of the proposed method to prior spectra, only the background spectral curve from scene 1 is used, while the object prior spectra are not included. In this case, due to the absence of object prior spectra, the target-domain guidance of the SSH becomes ineffective, and the generated hyperspectral representations no longer maintain spectral semantic constraints for the objects. The experimental results show that, without the guidance of object prior spectra, the performance of the proposed method decreases significantly. This observation further demonstrates the importance of prior spectra in guiding cross-domain object detection.

Regarding the selection of spectral priors, the background spectral prior is typically chosen from the background class

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TABLE VI
RECORDS OF COMPARATIVE EXPERIMENTS AND ABLATION EXPERIMENTS

Method	Detector Head	Reference	Dataset	mAP_{50}	AP_{ship}	AP_{car}	$AP_{long\ vehicle}$
AT	Faster RCNN	CVPR2022	SIMD2LWP2.0	14.9	12.7	21.3	29.0
CMT-AT	Faster RCNN	CVPR2023		16.6	12.1	24.1	13.6
MGADA	Faster RCNN	CVPR2022		21.0	19.1	21.0	19.9
PT	Faster RCNN	ICML2022		13.9	3.0	17.4	21.4
CMT-PT	Faster RCNN	CVPR2023		15.9	5.6	23.9	18.2
MRT	DETR	ICCV2023		19.4	12.7	16.5	29.0
DATR	DETR	TIP2025		22.9	9.5	25.0	34.1
Ours	Faster RCNN	\		**39.9**	**34.7**	**47.7**	**37.3**
w/o REM	Faster RCNN	\		22.5	21.4	22.7	23.4
w/o SHH	Faster RCNN			21.9	10.7	21.5	33.5
w/o heteroscedastic loss	Faster RCNN			37.7	33.8	43.7	35.5
w/o_spectral uncertainty loss	Faster RCNN			37.5	29.2	46.9	36.3
Method	Detector Head	Reference	Dataset	mAP_{50}	AP_{ship}	AP_{car}	$AP_{long\ vehicle}$
AT	Faster RCNN	CVPR2022	MVPOD2LWP2.0	13.3	9.7	27.9	2.4
CMT-AT	Faster RCNN	CVPR2023		13.9	8.9	15.5	17.3
MGADA	Faster RCNN	CVPR2022		7.8	12.5	8.5	2.6
PT	Faster RCNN	ICML2022		15.0	8.0	15.8	21.2
CMT-PT	Faster RCNN	CVPR2023		18.7	13.2	14.1	28.8
MRT	DETR	ICCV2023		15.2	10.1	3.4	**32.3**
DATR	DETR	TIP2025		24.3	18.4	33.6	20.9
Ours	Faster RCNN	\		**28.8**	**21.9**	**43.0**	21.6
Method	Detector Head	Reference	Dataset	mAP_{50}	mAP_{small}	mAP_{medium}	mAP_{large}
AT	Faster RCNN	CVPR2022	SIMD2M2SODAI	2.2	0.4	4.5	\
CMT-AT	Faster RCNN	CVPR2023		4.8	1.0	5.5	\
MGADA	Faster RCNN	CVPR2022		7.2	1.8	**9.7**	\
PT	Faster RCNN	ICML2022		2.9	0.5	3.6	\
CMT-PT	Faster RCNN	CVPR2023		2.4	0.6	3.5	\
MRT	DETR	ICCV2023		5.9	1.0	0.7	\
DATR	DETR	TIP2025		5.7	1.7	9.0	\
Ours	Faster RCNN	\		**14.6**	**3.9**	3.5	\

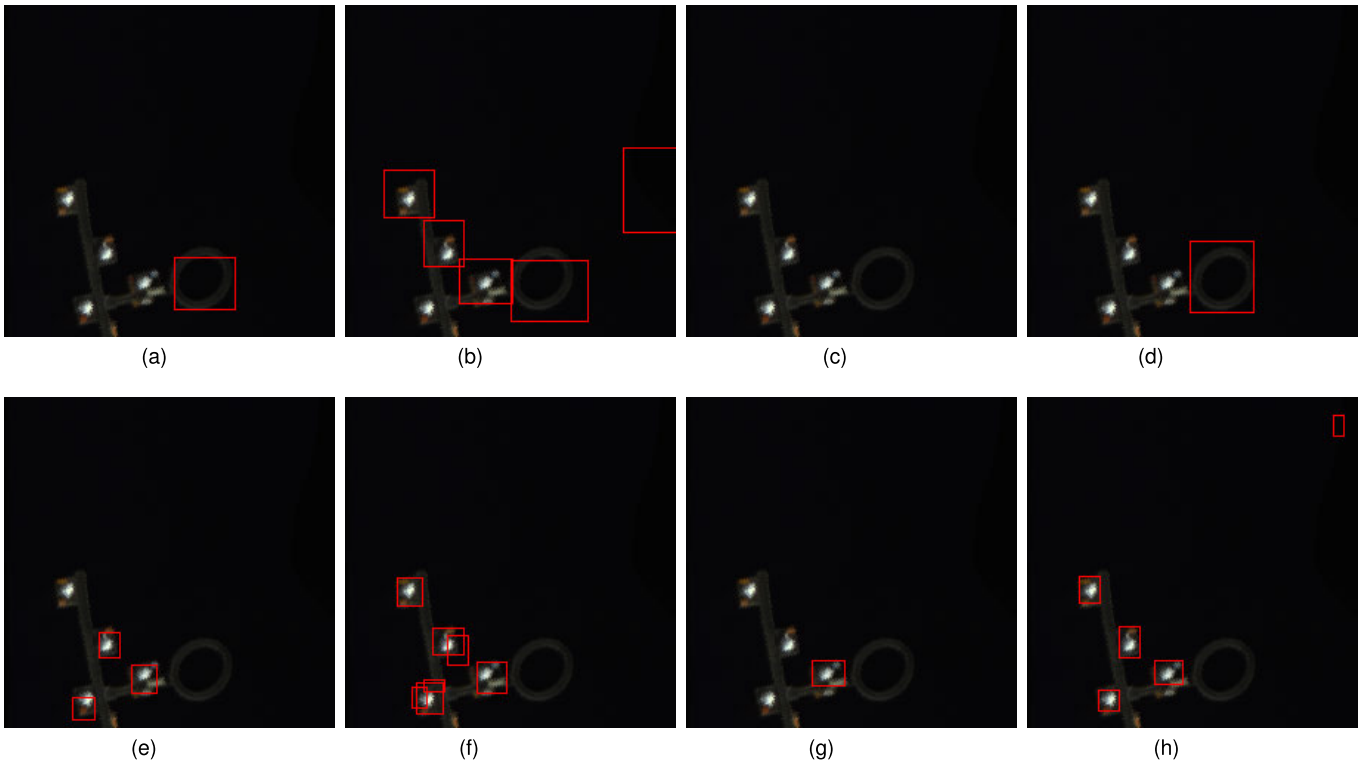


Fig. 11. Qualitative detection maps of different methods on SIMD2M2SODAI. (a) AT. (b) CMT-AT. (c) PT. (d) CMT-PT. (e) MRT. (f) MGADA. (g) DATR. (h) Ours.

that occupies the largest proportion of the overall dataset. In this dataset, the dominant background is seawater; therefore, the spectral signature of seawater is adopted as the background prior. In contrast, the selection of object spectral priors is

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TABLE VII

SENSITIVITY EXPERIMENT OF PRIOR SPECTRUM IN MVPOD2LWP2.0 SETTING. IT SHOULD BE NOTED THAT WHEN ONLY THE BACKGROUND SPECTRUM IS AVAILABLE, THE SEAWATER SPECTRUM FROM SCENE 1 IS USED

	mAP_{50}	AP_{ship}	AP_{car}	$AP_{longvehicle}$
Prior Scene 1	28.8	21.9	43.0	21.6
Prior Scene 2	27.9	19.8	48.4	15.4
Prior Scene 3	28.2	20.1	47.2	17.3
Only background prior	22.5	15.8	36.9	14.8

TABLE VIII

PARAMETER SENSITIVITY ANALYSIS OF RBF IN SIMD2M2SODAI

Fix $K = 48$			
	$mAP50$	AP_{small}	AP_{medium}
$\epsilon = 0.5$	13.7	3.6	6.9
$\epsilon = 1.0$	13.7	3.5	1.5
$\epsilon = 1.5$	14.6	3.9	3.5
$\epsilon = 2.0$	14.3	3.6	7.2
$\epsilon = 2.5$	15.3	4.0	5.8
Fix $\epsilon = 1.5$			
K=12	14.3	3.6	2.8
K=24	14.5	3.2	15.4
K=32	13.9	3.4	5.8
K=48	14.6	3.9	3.5
K=64	13.6	3.1	4.3

relatively flexible. The experimental results show that variations in object spectra across different scenes do not significantly affect the detection performance, with only minor fluctuations observed. Therefore, there is no strict rule for determining the object spectral priors in the proposed framework.

D. Parameter Sensitivity Analysis

To evaluate the sensitivity of the RBF parameters, related experiments are conducted under the SIMD2M2SODAI experimental setting by adjusting the two key parameters, K and ϵ. The corresponding results are reported in Table VIII. First, K is fixed at 48 while ϵ is varied across 0.5, 1.0, 1.5, 2.0, and 2.5. The results indicate that the performance is not significantly affected by different values of ϵ. Therefore, ϵ is set to the default value used in the previous comparative experiments, i.e., $\epsilon = 1.5$. Subsequently, with ϵ fixed, K is varied among 12, 24, 32, 48, and 64. The best performance is achieved when $K = 48$, which is consistent with the previously adopted default setting. Moreover, the performance degradation under other parameter settings is not substantial. These results demonstrate that the proposed method is not highly sensitive to the RBF parameters.

E. Ablation Study

To verify the effectiveness of the REM and SSH modules, we conduct ablation experiments on the SIMD2LWP2.0 scene, and the quantitative results are presented in Table VI. In the ablation study of the REM, we evaluate a variant that bypasses the REM and directly feeds the original images into the network. Specifically, for the target domain, the hyperspectral image is reduced to three bands using PCA before being input to the network. We find that all metrics show a noticeable decline. Moreover, removing the fusion with the spectral saliency map generated by the SSH and using only the spatial saliency map as input to the detection head also leads to a notable performance degradation, confirming the effectiveness of SSH and REM.

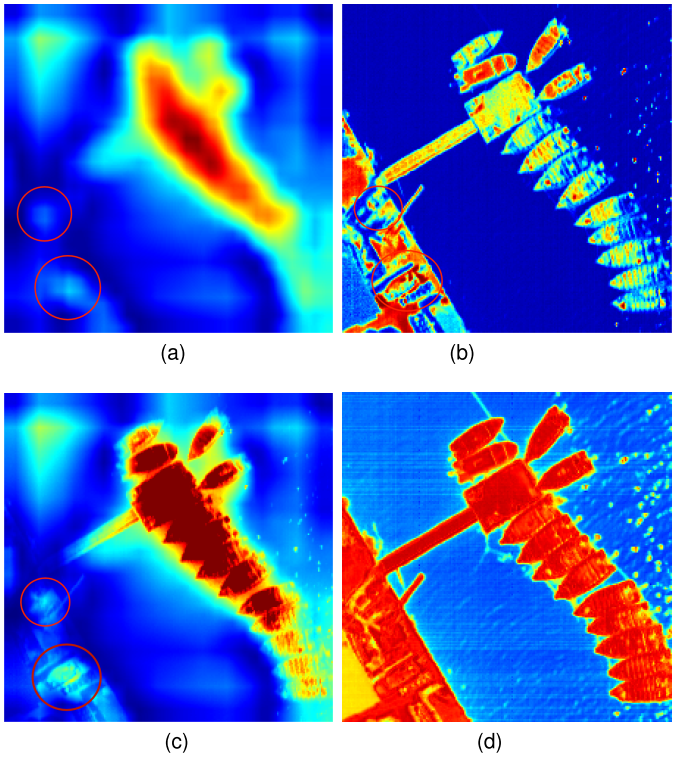


Fig. 12. Qualitative experiment on SSH. (a) P2 features of FPN. (b) Spectral saliency map generated by SSH. (c) Fused spatial–spectral saliency map. (d) Using only SAM without SSH.

To further demonstrate the effectiveness of the SSH module, we utilize EigenCAM [66] to visualize the spatial saliency map from FPN-P2, the spectral saliency map generated by SSH, and the spatial–spectral features obtained after their saliency-guided gated fusion. The corresponding visualizations are shown in Fig. 12. Through visualization, we observe that in the spatial saliency map of P2 as shown in Fig. 12(a), the spatial features of the two ships on the left are relatively weak, whereas in the spectral saliency map in Fig. 12(b), the corresponding ships exhibit stronger saliency. As shown in the fused features in Fig. 12(c), the fusion enhances the overall saliency of the objects. This raises the question of why the SAM method was not directly employed to generate the spectral saliency map, as opposed to using SSH. To address this, we performed comparative experiments utilizing the same spectral curves as in the primary experiments. The resulting SAM-based saliency map is presented in Fig. 12(d), where it is evident that all objects exhibiting similar spectral trends are highlighted. This leads to a negative impact on object detection performance.

To quantify the effectiveness of REM, we select two metrics: foreground activation ratio (FAR) and intersection over union (IoU). The FAR represents the proportion of pixels within the

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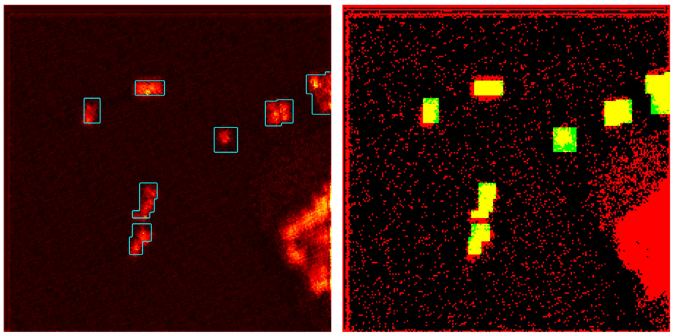


Fig. 13. Schematic of calculating IoU and FAR from the residual output by REM. The left side shows the residual image of the REM output, and the right side shows the corresponding binary image.

GT annotated object area that are activated by REM. A higher FAR value indicates a stronger response of REM at the target location. The IoU measures the ratio of the overlapping region between the REM activation area and the GT annotated area to the union of the two regions. A higher IoU value indicates better spatial alignment between the REM activation and the true object location. During the calculation, the top 20% of the largest values are selected as the targets to generate the corresponding binary map. Then, IoU and FAR are computed. For example, in Fig. 13, which presents a relatively dense object scenario, an IoU of 0.3504 and an FAR of 0.5842 are obtained. This indicates that the residuals output by REM align well with the relatively densely distributed small objects.

To verify the effectiveness of the losses introduced in the proposed method, ablation experiments were conducted on the heteroscedastic loss and the spectral uncertainty loss, and the corresponding results are reported in Table VI. The experimental results show that removing either the heteroscedastic loss or the spectral uncertainty loss leads to an approximate 2% *mAP* degradation. Moreover, when the spectral uncertainty loss is removed, the performance of the ship category, which constitutes the majority class in the LWP2.0 dataset, drops by approximately 5%. These results demonstrate that both loss terms play an important role in the proposed framework.

F. Discussion

To demonstrate the advantage of REM in extracting shared spatial features between domains, we visualize the original and residual images separately in the source and target domains using the t-SNE [67] method. Fig. 14(a) and (b) shows the t-SNE visualization of features from the original inputs and the residual representations obtained by the proposed REM. The original features exhibit a large domain discrepancy between VLRSI and HRSI samples, with scattered and irregular distributions. In contrast, the residual features become more compact within each domain and display more structured and symmetric distributions across domains. This demonstrates that the REM effectively suppresses domain-specific background variations and enhances domain-shared semantic cues, providing a more discriminative and transferable feature space for subsequent spectral–spatial alignment. In addition, the residual maps generated by the REM are visualized in

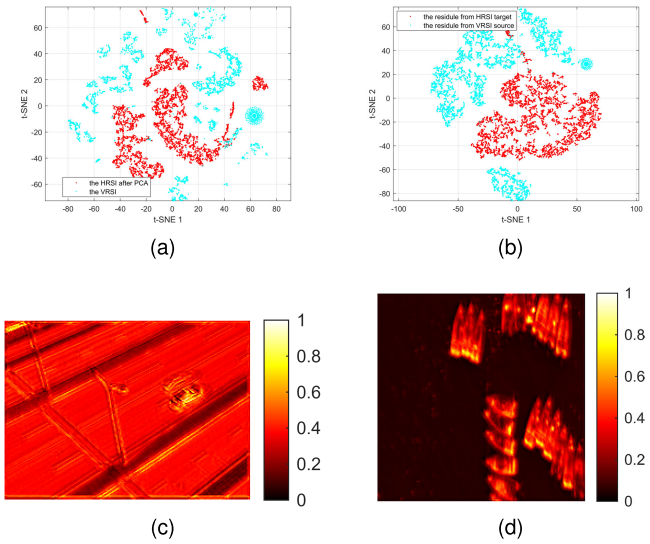


Fig. 14. (a) Schematic of the original image distribution in the source domain and the three-band image distribution in the target domain after PCA. (b) Schematic of the source-domain residual distribution and target-domain residual distribution generated by REM. (c) Residual results of the source-domain dataset—MVPOD. (d) Residual results of target-domain dataset—LWP2.0.

Fig. 14(c) and (d). As shown, the residuals effectively highlight the objects of interest in both the source and target domains, thereby demonstrating their effectiveness.

Here, the applicability boundaries of the proposed method are discussed. The method is particularly suitable for scenarios where the overall dataset exhibits relatively smooth backgrounds and where target objects are relatively small and sparsely distributed, which is a common characteristic of many remote sensing detection datasets. Nevertheless, the proposed approach remains effective in moderately dense object scenarios, as verified in the additional experiments on the M2SODAI dataset. This robustness can be attributed to two main reasons. First, the REM emphasizes foreground regions by reconstructing the dominant background structures. Since background regions usually occupy the majority of pixels even in moderately dense scenes, the reconstruction process still focuses primarily on background patterns, allowing residual features to highlight object regions. Second, relatively smooth backgrounds generally imply more homogeneous background materials. Under such conditions, the proposed method can more effectively capture the spectral characteristics of the background, thereby enhancing the discriminability between objects and the background. In contrast, highly textured backgrounds may introduce complex spatial patterns that are difficult to reconstruct accurately, which may weaken the foreground–background separability of the residual representations.

V. Conclusion

To achieve cross-domain object detection from VLRSI to HRSI, we propose VHCDN. An REM is first introduced to align the spatial features between VLRSI and HRSI, which is based on the principle that objects occupy a much smaller

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proportion than the background in remote sensing images. To model spectral uncertainty, a precollected HRSI is introduced into the source-domain datasets to guide the generation of reasonable spectral curves. Subsequently, we propose the SSH, which employs a spectral heteroscedasticity loss and a spectral uncertainty loss to transfer the learned spectral uncertainty from the source domain to the target domain. Under three experimental settings, our method achieves the best performance, and the effectiveness of both the REM and SSH modules is validated.

In the future, to more fully exploit the advantages of hyperspectral imagery, we plan to investigate a unified framework that integrates object-level detection with pixel-level analysis. Such integration is expected not only to localize targets of interest at the object level, but also to characterize intraobject spectral variability, thereby identifying spectral discrepancies within objects belonging to the same semantic category. This capability would enable a more fine-grained understanding of material composition and surface properties, and could be further extended to practical monitoring scenarios, such as detecting ships carrying or composed of anomalous materials at sea.

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